This operation requires linear time in the worst case, as we might have to update up to $n-1$ elements (where n is the total number of elements in the universe U).

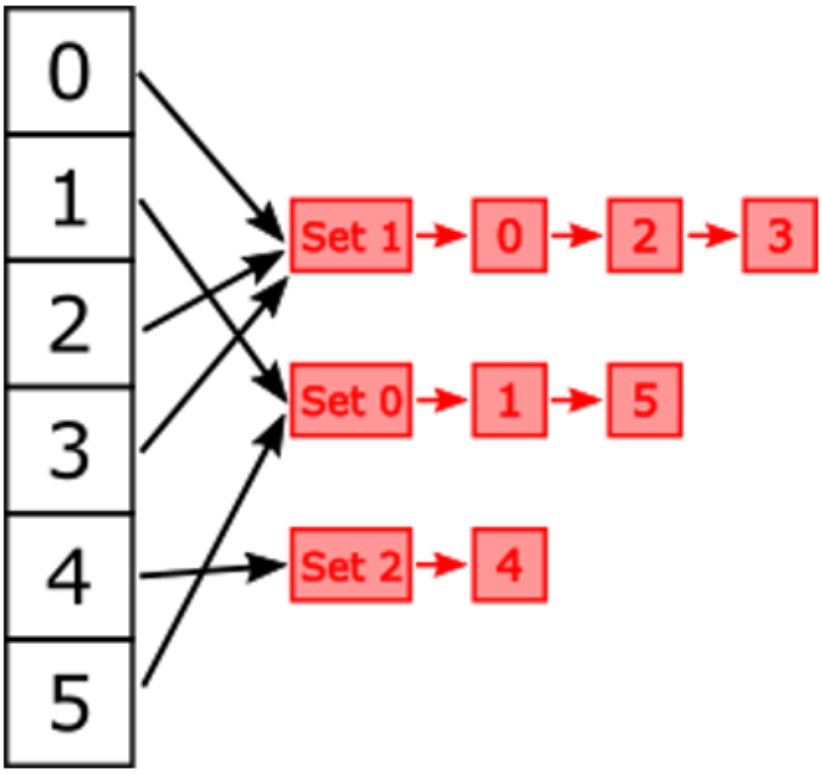


Figure 5.4 Representing a disjoint set with lists. Each array element stores a pointer to the head a linked list; each linked list, in turn, represents a set. Here sets are numbered arbitrarily, as the index doesn’t really provide any information on the set (nor can it be retrieved).

There is one way to slightly improve the expected number of assignments we have to perform: by always appending the shortest of the two lists, we will make sure that we won't have to update more than $n/2$ elements’ pointers.